In [ ]:
```
 Experiment 4: Data Manipulation with Pandas
PIN:241UDS0101

Aim
Practice selecting, calculating, filtering, sorting, updating, and renaming
data in a retail sales DataFrame.
```

In [ ]:
```
## Step 1 – Install and Import Pandas Library
```

In [4]:
```
import pandas as pd
```

In [ ]:
```
## Step 2 – Importing the Dataset
Keep the CSV file in the same folder as this notebook.
```

In [13]:
```
df_241UDS0101 = pd.read_csv('Experiment_4_241UDS0101_Retail_Sales_Data.csv')
df_241UDS0101.head(5)
```

Out[13]:

Columns 1–8 of 11

| | Product_ID | Product_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent |
|---|---|---|---|---|---|---|---|
| 0 | P1001 | Wireless Mouse | Electronics | Vijayawada | 79 | 1838 | 15.0 |
| 1 | P1002 | Notebook Set | Stationery | Hyderabad | 44 | 1034 | 5.0 |
| 2 | P1003 | Water Bottle | Home | Delhi | 96 | 256 | 10.0 |
| 3 | P1004 | USB-C Cable | Accessories | Chennai | 63 | 1034 | 6.0 |
| 4 | P1005 | Desk Lamp | Electronics | Chennai | 96 | 210 | 6.0 |

Columns 9–11 of 11 · first column repeated

| | Customer_Rating | Stock_Available | Payment_Method |
|---|---|---|---|
| 0 | 4.0 | 36 | UPI |
| 1 | 4.1 | 53 | Cash |
| 2 | 4.9 | 46 | Cash |
| 3 | 3.6 | 61 | UPI |
| 4 | 4.5 | 48 | Cash |

In [7]:
```
## Step 3 – Set Index for the DataFrame
```

In [14]:
```
df_241UDS0101 = df_241UDS0101.set_index('Product_ID')
df_241UDS0101.head()
```

Out[14]: Columns 1–8 of 10

| | Product_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **Product_ID** | | | | | | | |
| **P1001** | Wireless Mouse | Electronics | Vijayawada | 79 | 1838 | 15.0 | 4.0 |
| **P1002** | Notebook Set | Stationery | Hyderabad | 44 | 1034 | 5.0 | 4.1 |
| **P1003** | Water Bottle | Home | Delhi | 96 | 256 | 10.0 | 4.9 |
| **P1004** | USB-C Cable | Accessories | Chennai | 63 | 1034 | 6.0 | 3.6 |
| **P1005** | Desk Lamp | Electronics | Chennai | 96 | 210 | 6.0 | 4.5 |

Columns 9–10 of 10 · first column repeated

| | Stock_Available | Payment_Method |
|---|---|---|
| **Product_ID** | | |
| **P1001** | 36 | UPI |
| **P1002** | 53 | Cash |
| **P1003** | 46 | Cash |
| **P1004** | 61 | UPI |
| **P1005** | 48 | Cash |

In [ ]:
```
## Step 4 – Info of the DataFrame
```

In [15]:
```
df_241UDS0101.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 30 entries, P1001 to P1030
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Product_Name      30 non-null     object 
 1   Category          30 non-null     object 
 2   City              30 non-null     object 
 3   Units_Sold        30 non-null     int64  
 4   Unit_Price        30 non-null     int64  
 5   Discount_Percent  30 non-null     float64
 6   Customer_Rating   30 non-null     float64
 7   Stock_Available   30 non-null     int64  
 8   Payment_Method    30 non-null     object 
dtypes: float64(2), int64(3), object(4)
memory usage: 2.3+ KB
```

In [ ]:
```
## Step 5 — Columns & Rows Selection
```

In [16]:
```
df_241UDS0101['Product_Name']
```

Out[16]:
```
Product_ID
P1001       Wireless Mouse
P1002         Notebook Set
P1003         Water Bottle
P1004          USB-C Cable
P1005            Desk Lamp
P1006             Pen Pack
P1007    Bluetooth Speaker
P1008           Coffee Mug
P1009             Backpack
P1010             Keyboard
P1011          File Folder
P1012      Table Organizer
P1013          Phone Stand
P1014     Highlighter Pack
P1015           Power Bank
P1016            Lunch Box
P1017         Sticky Notes
P1018        Laptop Sleeve
P1019            LED Strip
P1020        Cushion Cover
P1021           Marker Set
P1022            Earphones
P1023           Wall Clock
P1024         Travel Pouch
P1025           Calculator
P1026           Smart Plug
P1027          Storage Box
P1028      Document Folder
P1029          Phone Pouch
P1030             Mini Fan
Name: Product_Name, dtype: object
```

In [ ]:
```
## Step 6 — Selecting Multiple Columns
```

In [17]:
```
df_241UDS0101[[ 'Product_Name', 'Category', 'City' ]].head(3)
```

Out[17]:

| Product_ID | Product_Name | Category | City |
|---|---|---|---|
| **P1001** | Wireless Mouse | Electronics | Vijayawada |
| **P1002** | Notebook Set | Stationery | Hyderabad |
| **P1003** | Water Bottle | Home | Delhi |

In [ ]:
```
## Step 7 — Select Row Using iloc
```

In [18]:
```
df_241UDS0101.iloc[1]
```

Out[18]:
```
Product_Name        Notebook Set
Category              Stationery
City                   Hyderabad
Units_Sold                    44
Unit_Price                  1034
Discount_Percent             5.0
Customer_Rating              4.1
Stock_Available               53
Payment_Method              Cash
Name: P1002, dtype: object
```

In [ ]:
```
## Step 8 – Select Row Using loc
```

In [19]:
```
df_241UDS0101.loc['P1030']
```

Out[19]:
```
Product_Name          Mini Fan
Category            Stationery
City                 Bangalore
Units_Sold                  96
Unit_Price                1248
Discount_Percent          15.0
Customer_Rating            5.0
Stock_Available             48
Payment_Method            Cash
Name: P1030, dtype: object
```

In [ ]:
```
## Step 9 – Display Initial Rows
```

In [20]:
```
df_241UDS0101.head(2)
```

Out[20]:

Columns 1–8 of 10

| | Product_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **Product_ID** | | | | | | | |
| **P1001** | Wireless Mouse | Electronics | Vijayawada | 79 | 1838 | 15.0 | 4.0 |
| **P1002** | Notebook Set | Stationery | Hyderabad | 44 | 1034 | 5.0 | 4.1 |

Columns 9–10 of 10 · first column repeated

| | Stock_Available | Payment_Method |
|---|---|---|
| **Product_ID** | | |
| **P1001** | 36 | UPI |
| **P1002** | 53 | Cash |

In [ ]:
```
## Step 10 – Creating Total Sales Column
```

In [21]:
```
df_241UDS0101['Total_Sales_241UDS0101'] = df_241UDS0101['Units_Sold'] *
df_241UDS0101['Unit_Price']
df_241UDS0101.head(2)
```

Out[21]: Columns 1–8 of 11

| | Product_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **Product_ID** | | | | | | | |
| **P1001** | Wireless Mouse | Electronics | Vijayawada | 79 | 1838 | 15.0 | 4.0 |
| **P1002** | Notebook Set | Stationery | Hyderabad | 44 | 1034 | 5.0 | 4.1 |

Columns 9–11 of 11 · first column repeated

| | Stock_Available | Payment_Method | Total_Sales_241UDS0101 |
|---|---|---|---|
| **Product_ID** | | | |
| **P1001** | 36 | UPI | 145202 |
| **P1002** | 53 | Cash | 45496 |

In [ ]:
```
## Step 11 – Creating Discount Amount Column
```

In [22]:
```
df_241UDS0101['Discount_Amount_241UDS0101'] =
df_241UDS0101['Total_Sales_241UDS0101'] * df_241UDS0101['Discount_Percent']
/ 100
df_241UDS0101.head(2)
```

Out[22]: Columns 1–8 of 12

| | Product_Name | Category | City | Units_Sold | Unit_Price | Discount_Percent | Customer_Rating |
|---|---|---|---|---|---|---|---|
| **Product_ID** | | | | | | | |
| **P1001** | Wireless Mouse | Electronics | Vijayawada | 79 | 1838 | 15.0 | 4.0 |
| **P1002** | Notebook Set | Stationery | Hyderabad | 44 | 1034 | 5.0 | 4.1 |

Columns 9–12 of 12 · first column repeated

| | Stock_Available | Payment_Method | Total_Sales_241UDS0101 | Discount_Amount_241UDS0101 |
|---|---|---|---|---|
| **Product_ID** | | | | |
| **P1001** | 36 | UPI | 145202 | 21780.3 |
| **P1002** | 53 | Cash | 45496 | 2274.8 |

In [ ]:
```
## Step 12 – Creating Net Sales Column
```

In [ ]:
```
df_241UDS0101['Net_Sales_241UDS0101'] =
df_241UDS0101['Total_Sales_241UDS0101'] -
df_241UDS0101['Discount_Amount_241UDS0101']
df_241UDS0101.head(2)
```

In [ ]:
```
## Step 13 – Round Calculated Values
```

In [ ]:
```
df_241UDS0101['Discount_Amount_241UDS0101'] =
df_241UDS0101['Discount_Amount_241UDS0101'].round(2)
df_241UDS0101['Net_Sales_241UDS0101'] =
df_241UDS0101['Net_Sales_241UDS0101'].round(2)
df_241UDS0101.head(2)
```

In [ ]:
```

## Step 14 – Conditional Filtering: Units Sold Greater Than 50
```

In [ ]:
```
df_241UDS0101[df_241UDS0101['Units_Sold'] > 50].head(5)
```

In [ ]:
```
## Step 15 – Products with High Sales and Low Stock
```

In [ ]:
```
df_241UDS0101[(df_241UDS0101['Total_Sales_241UDS0101'] > 30000) &
(df_241UDS0101['Stock_Available'] < 25)].head(5)
```

In [ ]:
```
## Step 16 – Products from Hyderabad OR Mumbai
```

In [ ]:
```
df_241UDS0101[df_241UDS0101['City'].isin(['Hyderabad', 'Mumbai'])].head()
```

In [ ]:
```
## Step 17 – Rating ≥ 4.5 AND Stock < 30
```

In [ ]:
```
df_241UDS0101[(df_241UDS0101['Customer_Rating'] ⩾ 4.5) &
(df_241UDS0101['Stock_Available'] < 30)]
```

In [ ]:
```
## Step 18 – Filtering Multiple Cities Using .isin()
```

In [ ]:
```
df_241UDS0101[df_241UDS0101['City'].isin(['Hyderabad', 'Bangalore',
'Pune'])].head()
```

In [ ]:
```
## Step 19 – Filtering Multiple Categories Using .isin()
```

In [ ]:
```
df_241UDS0101[df_241UDS0101['Category'].isin(['Electronics',
'Home'])].head()
```

In [ ]:
```
## Step 20 – Filtering Multiple Payment Methods Using .isin()
```

In [ ]:
```
df_241UDS0101[df_241UDS0101['Payment_Method'].isin(['UPI', 'Card'])].head()
```

In [ ]:
```
## Step 21 – Sorting by Units Sold (Descending)
```

In [ ]:
```
df_241UDS0101.sort_values('Units_Sold', ascending=False).head()
```

In [ ]:
```
## Step 22 – Sort by Unit Price (Ascending)
```

In [ ]:
```
df_241UDS0101.sort_values('Unit_Price', ascending=True).head()
```

In [ ]:
```
## Step 23 – Sorting Using Multiple Columns
```

In [ ]:
```
df_241UDS0101.sort_values(['Category', 'Units_Sold'], ascending=[True,
False]).head()
```

In [ ]:
```
## Step 24 – Updating Data in the Table
```

In [ ]:
```
df_241UDS0101['City'] = df_241UDS0101['City'].replace('Hyderabad', 'Hyd')
df_241UDS0101[df_241UDS0101['City'] == 'Hyd'].head(2)
```

In [ ]:
```
## Step 25 – Updating a Specific Product Value
```

In [ ]:
```
df_241UDS0101.loc['P1001', 'Unit_Price'] = 700
df_241UDS0101.loc['P1001', 'Unit_Price']
```

In [ ]:
```
## Step 26 – Changing Column Names
```

In [ ]:
```
df_241UDS0101 = df_241UDS0101.rename(columns={'Product_Name': 'Item_Name',
'City': 'Location', 'Customer_Rating': 'Rating', 'Net_Sales_241UDS0101':
'Revenue'})
df_241UDS0101.head(1)
```

In [ ]:
```
## Step 27 – Final DataFrame
```

In [ ]:
```
df_241UDS0101.head()
```

In [ ]:
```
## Step 28 – Final Dataset Information
```

In [ ]:
```
df_241UDS0101.info()
```

In [ ]:

```
 Result :-
Pandas operations were used to select rows and columns, create calculated
columns, filter and sort records, update values, and rename columns.
```